MARKING SCHEME

Model Paper — 04

Mathematics I

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Note (for teachers):

- Part A: 2 marks per question (1 for the correct method / working, 1 for the correct answer). Total 50 marks.
- Part B: 10 marks per question. Total 50 marks. The marks for each step are shown in the right-hand column and add up to the marks printed on the question paper.
- Award full marks for any correct alternative method. Accept answers within the stated ranges for values read from graphs. Penalise a missing or wrong unit only once in a question.

Part A — Answers

1. $12 \times 10 = 120$ worker-days; workers needed for 6 days $= 120 \div 6 = 20$; extra workers $= 20 - 12$

Final answer: 8 workers

(2 marks)

2. Through E draw a line parallel to AB: $\angle BAE + \angle ECD = \angle AEC$, so $\angle BAE = 100° - 35°$

Final answer: 65°

(2 marks)

3. LCM of the denominators = 12x: $\frac{5 \times 6 + 7 \times 4 - 1 \times 3}{12x} = \frac{30 + 28 - 3}{12x}$

Final answer: $\frac{55}{12x}$

(2 marks)

4. The faces of a right triangular prism are 2 triangles and 3 rectangles (a square is a special rectangle); a rhombus that is not a square cannot occur.

Final answer: (ii)

(2 marks)

5. $20° \leftrightarrow 30$ kg, so $1° \leftrightarrow 1.5$ kg; angle of R $= 360° - 180° - 20° = 160°$; quantity $= 160 \times 1.5$

Final answer: 240 kg

(2 marks)

6. ADBE is cyclic: $\angle ADB = 180° - 124° = 56°$. DC passes through the centre and the midpoint of AB, so DC $\perp$ AB and DA = DB; DC bisects $\angle ADB$: $\angle CDB = 56° \div 2$

Final answer: 28°

(2 marks)

7. $5(x^2 - 9) = 5(x^2 - 3^2)$

Final answer: $5(x + 3)(x - 3)$

(2 marks)

8. $\lg 70 = \lg (7 \times 10) = \lg 7 + \lg 10 = 0.8451 + 1$

Final answer: 1.8451

(2 marks)

9. a is marked between the dashed horizontal through B and BC (below it); b is marked at C between CA (ground) and CB. The dashed horizontal through B $\parallel$ AC, so a and b are alternate angles.

Final answer: a = b (alternate angles)

(2 marks)

10. h = 6r: curved surface : base = $2\pi r(6r) : \pi r^2 = 12\pi r^2 : \pi r^2$

Final answer: 12 : 1

(2 marks)

11. Total outcomes = $4 \times 4 = 16$; sum 10: (2, 8), (8, 2), (5, 5) $\rightarrow$ 3 outcomes

Final answer: $\frac{3}{16}$

(2 marks)

12. BA = BD: $\angle BDA = 40°$, so $\angle DBC = 40° + 40° = 80°$ (exterior angle). DB = DC: $\angle DCB = 80°$. In $\triangle ADC$: $x = 180° - 40° - 80°$

Final answer: 60°

(2 marks)

13. A = {2, 3, 5, 7}, B = {1, 3, 5, 7, 9}; A $\cup$ B = {1, 2, 3, 5, 7, 9} (n = 6), A $\cap$ B = {3, 5, 7}

Final answer: (ii)

(2 marks)

14. $\frac{24x^4y^5}{8x^2y}$

Final answer: $3x^2y^4$

(2 marks)

15. $r = (-18) \div (-6) = 3$; $T_5 = T_3 \times r^2 = -18 \times 9$

Final answer: −162

(2 marks)

16. $\angle BDC = \angle BAC = 41°$ (angles in the same segment); $\angle ADB = \angle BDC = 41°$, so $\angle ADC = 82°$. $\angle CBE = \angle ADC$ (exterior angle of a cyclic quadrilateral)

Final answer: 82°

(2 marks)

17. HCF of 16, 24, 40 = 8; lowest power of x = x; lowest power of y = y

Final answer: 8xy

(2 marks)

18. $\pi r^2 h = 3850 \times h$, so $\pi r^2 = 3850$; $r^2 = 3850 \times \frac{7}{22} = 1225$

Final answer: r = 35 cm

(2 marks)

19. DB = AD = 3 cm; BC = 2 × DE = 8 cm (midpoint theorem); $AE = \sqrt{(3^2 + 4^2)} = 5$ cm, so EC = AE = 5 cm. Perimeter = 3 + 8 + 5 + 4

Final answer: 20 cm

(2 marks)

20. Gradient = $\frac{11 - 5}{4 - 2} = 3$; $y - 5 = 3(x - 2)$

Final answer: y = 3x − 1

(2 marks)

21. $\angle DAE = \angle BCD = 70°$ (opposite angles of a parallelogram); AE is a diameter, so $\angle ADE = 90°$; $\angle AED = 180° - 90° - 70° = 20°$; $\angle DEF = 180° - 20°$

Final answer: 160°

(2 marks)

22. Amount at the end of year 1 = 44 100 − 2100 = 42 000; $P \times \frac{105}{100} = 42\,000$, so $P = 42\,000 \times \frac{100}{105}$

Final answer: Rs. 40 000

(2 marks)

23. CB = CD, so $\angle CBD = (180° - 40°) \div 2 = 70°$; AC ∥ BD gives $\angle ACB = \angle CBD = 70°$ (alternate); $\angle AEB = ½\ \angle ACB$

Final answer: 35°

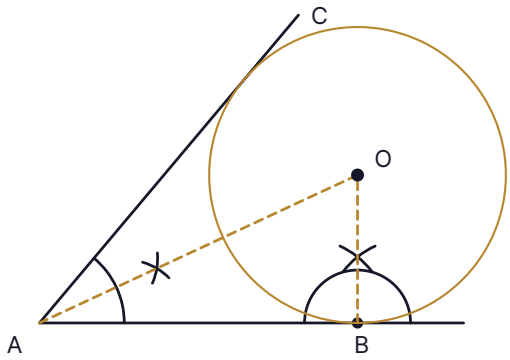
(2 marks)

24. $4x > 2 - 23 = -21$, so $x > -21/4$ (= −5¼)

Final answer: −5, −4, −3, −2, −1

(2 marks)

25. Bisect ∠BAC (join A to the point where the two arcs cross) and draw the perpendicular to AB at B (join B to the point where its arcs cross, and extend); the two lines meet at the centre O. (1 mark: bisector of ∠BAC completed; 1 mark: perpendicular at B completed and O marked at the intersection.)



Final answer: O = intersection of the angle bisector at A and the perpendicular at B

(2 marks)

Part B — Answers and mark allocation

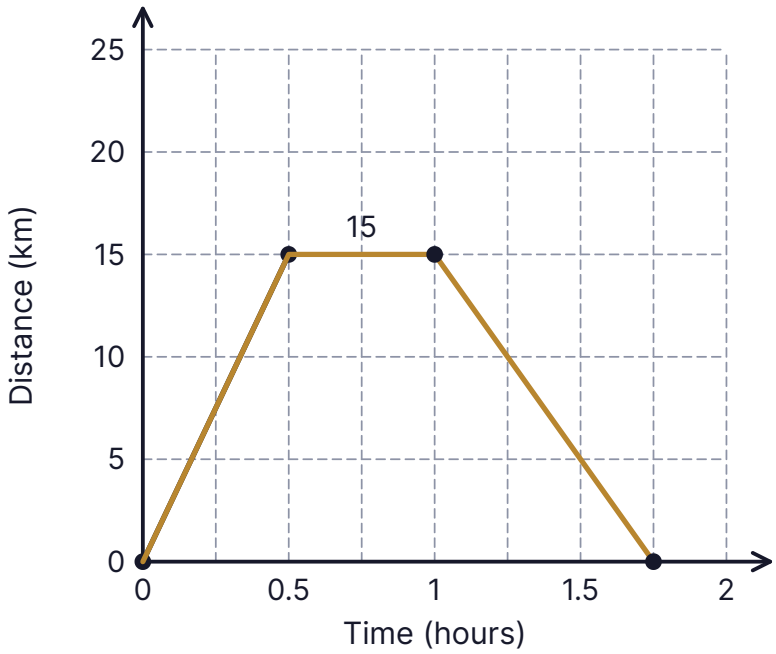
Question B1

(i)	
Remaining after food = $1 - \frac{1}{8} = \frac{7}{8}$	**1**
(ii)	
Spent on other expenses = $\frac{2}{7} \times \frac{7}{8} = \frac{1}{4}$ of the money received (or remainder = $\frac{5}{7}$ of $\frac{7}{8}$)	**1**
Fraction remaining = $\frac{7}{8} - \frac{1}{4} = \frac{5}{8}$	**1**
(iii)	
$\frac{5}{8}$ of the money received = 72 000, so money received = $72\,000 \times \frac{8}{5}$ = Rs. 115 200	**1**
Food = $\frac{1}{8} \times 115\,200$ = Rs. 14 400; other expenses = $\frac{1}{4} \times 115\,200$ = Rs. 28 800	**1**
(iv)	
Money received = Rs. 60 000 + 92% of the part of the salary above Rs. 60 000	**1**
$115\,200 - 60\,000 = 55\,200$ = 92% of the excess, so excess = $55\,200 \times \frac{100}{92}$ = Rs. 60 000	**1**
Salary = 60 000 + 60 000 = Rs. 120 000	**1**
(v)	
Food as a percentage of salary = $\frac{14\,400}{120\,000} \times 100$	**1**
= 12%	**1**

Total marks: 10

Question B2

(a)(i) — completed graph shown below	
Horizontal line from (0.5, 15) to (1, 15) (stay of 30 minutes)	**1**
Point (1.75, 0) correctly plotted (1 h + 45 min = 1.75 h)	**1**
Straight line from (1, 15) to (1.75, 0) joined	**1**
(a)(ii)	
Speed = distance ÷ time = 15 ÷ 0.75 (45 min = 0.75 h)	**1**
= 20 km h^{-1}	**1**
(b)(i)	
Length of arc AEB = $\frac{1}{2} \times 2 \times \frac{22}{7} \times 7 = 22$ cm	**1**
Perimeter = 22 + 22 + 14 + 22 = 80 cm	**1**
(b)(ii)	
Area of rectangle = 14 × 22 = 308 cm^2	**1**
Area of semicircle = $\frac{1}{2} \times \frac{22}{7} \times 7 \times 7 = 77$ cm^2	**1**
308 ÷ 77 = 4 times	**1**



Completed graph: (0, 0) → (0.5, 15) → (1, 15) → (1.75, 0). Accept points within half a small square.

Total marks: 10

Question B3

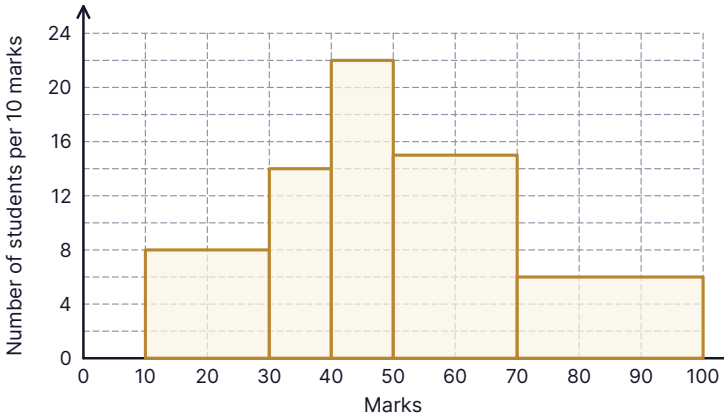
(i)	
Rates = 120 000 × $\frac{15}{100}$	**1**
= Rs. 18 000	**1**
(ii)	
Annual rent = 15 000 × 12 = Rs. 180 000; maintenance = 180 000 × $\frac{10}{100}$ = Rs. 18 000	**1**
Remaining = 180 000 − 18 000 − 18 000 = Rs. 144 000	**1**
(iii)(a)	
Number of shares = 10 800 ÷ 3 (dividend per share)	**1**
= 3600 shares	**1**
Total invested = 3600 × 50 = Rs. 180 000	**1**
(iii)(b)	
Extra money added = 180 000 − 144 000 = Rs. 36 000	**1**
Percentage = $\frac{36\,000}{180\,000}$ × 100	**1**
= 20%	**1**

Total marks: 10

Question B4

(i)	
Number of students = height × (class width ÷ 10)	**1**
30 – 40: 14 and 40 – 50: 22	**1**
50 – 70: 15 × 2 = 30 and 70 – 100: 6 × 3 = 18	**1**
(ii)	
Bars drawn with no gaps on the class boundaries 10, 30, 40, 50, 70, 100; heights of the first two bars 8 and 14	**1**
Heights of the third and fourth bars 22 and 15	**1**
Height of the last bar 6 (bars wider for wider classes)	**1**
(iii)	
Cumulative frequencies 16, 30, 52, 82, 100 (all correct 2 marks; at least 3 correct 1 mark)	**2**
(iv)	
Students with 50 or more marks = 30 + 18 = 48 (or 100 − 52)	**1**
Probability = $\frac{48}{100} = \frac{12}{25}$	**1**

Marks	Number of students	Cumulative frequency
10 – 30	16	16
30 – 40	14	30
40 – 50	22	52
50 – 70	30	82
70 – 100	18	100

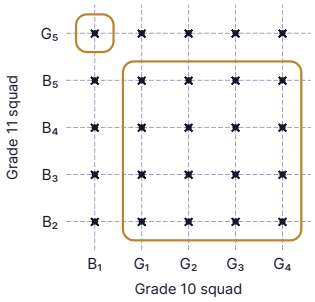


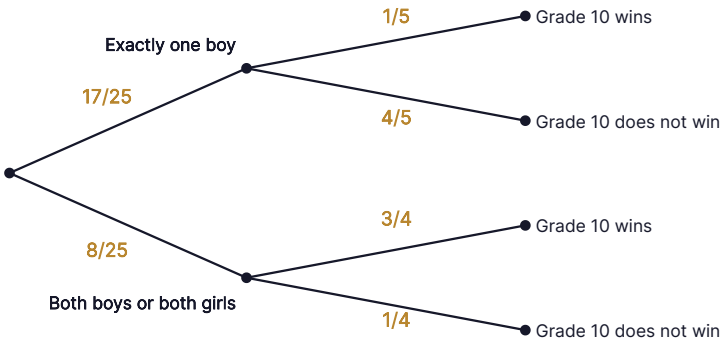
Histogram: bar heights 8, 14, 22, 15, 6 (students per 10 marks).

Total marks: 10

Question B5

(i)	
All 25 grid points marked with "×" (5 columns × 5 rows)	**2**
(ii)	
Closed curves around the 1 point (B_1 with G_5) and the 16 points (G_1, G_2, G_3, G_4 with B_2, B_3, B_4, B_5)	**1**
Number of favourable outcomes = 1 + 16 = 17	**1**
Probability = $\frac{17}{25}$	**1**
(iii)	
First branches: $\frac{17}{25}$ (exactly one boy) and $\frac{8}{25}$ (both boys or both girls: 4 + 4 = 8 outcomes)	**1**
Second branches under "exactly one boy": $\frac{1}{5}$ (wins), $\frac{4}{5}$ (does not win)	**1**
Second branches under "both boys or both girls": $\frac{3}{4}$ (wins), $\frac{1}{4}$ (does not win)	**1**
(iv)	
P(does not win) = $\frac{17}{25} \times \frac{4}{5} + \frac{8}{25} \times \frac{1}{4} = \frac{68}{125} + \frac{10}{125}$	**1**
= $\frac{78}{125}$	**1**





Total marks: 10